

Sangram Rout

London, UK · Right to work in the UK — no sponsorship required (HPI visa)

Robotics & AI Engineer · Reinforcement Learning · Humanoid Control · Robot Policy Learning

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SUMMARY

Robotics & AI Engineer specialising in reinforcement learning for humanoid locomotion, world models, and robot policy learning — combining rare hands-on mechanical engineering experience with end-to-end ML implementation.

Re-implemented 12+ SOTA papers from scratch (DreamerV4, Pi0, Genie, S2GO, CarPlanner, TopoDiffuser).

Published in Nature, awarded a U.S. patent, and co-PI on a \$1.3M DOE project.

EDUCATION

M.S. Mechanical Engineering (Robotics Focus) · Northwestern University, Evanston, IL

2019 – 2021

Thesis: Soft Robotics Design & Simulation · Coursework: RL, Control Systems, Robot Manipulation, Machine Dynamics

PUBLICATIONS & PATENTS

- ▶ Nature Scientific Reports (2022): KnitMorphs — Numerical Simulation of Knit-Based Morphable Structures.
- ▶ U.S. Patent (approved): Systems & Methods for Dieless Composite Forming.

SKILLS

RL & Control	PPO, SAC, TD-MPC, AMP / Motion Imitation, Reward Shaping, Multi-Agent RL, Human-in-the-Loop
Simulation	IsaacSim / Isaac Lab, MuJoCo, Gazebo, ROS2 (Nav2, SLAM, URDF), 3D Gaussian Splatting
ML / AI	PyTorch, Stable Baselines3, World Models (DreamerV4, Genie), VLA (Pi0, Dot Policy), Diffusion Models, VLMs
Languages	Python, C++ (basics), ONNX model compression, Git, Arduino / Embedded C
Mechanical	SolidWorks, Ansys FEA/Thermal, GD&T, ASME/EN standards, PCB design
Certifications	NVIDIA GTC 2025 — AI Agents with Multimodal Models · ROS2 Nav2 · Kalman Filter / Sensor Fusion (Coursera)

ROBOTICS & AI PROJECTS

1 · World Models & Scene Understanding

Humanoid World Model — End-to-End Reimplementation [↗](#)

Reimplemented a humanoid world model from scratch: latent dynamics model, action-conditioned prediction head, and rollout evaluation. Verified the model predicts multi-step joint trajectories accurately enough to support policy training in imagination.

Stack: World model · latent dynamics · PyTorch · humanoid sim

DreamerV4 — Open-Source Reimplementation [↗](#)

Reimplemented DreamerV4 on a scaled-down MineRL dataset to fit a single GPU. Verified RSSM latent dynamics, imagination-based actor-critic training, and KL balancing against the paper.

Stack: *DreamerV4 · RSSM · world model · MineRL · PyTorch*

Genie — Generative Interactive Environments [↗](#)

Implemented Genie's three components — video tokeniser, latent action model, and spatiotemporal dynamics model — and validated training and generation on a PONG dataset.

Stack: *Genie · latent action model · video generation · spatiotemporal transformer · PyTorch*

CoPilot4D — LiDAR Future Prediction (In Progress) [↗](#)

Reimplementing CoPilot4D's LiDAR world model for driving: the 4D occupancy representation and masked-autoencoder pre-training, targeting 3-second point-cloud forecasts without camera input.

Stack: *CoPilot4D · LiDAR · 4D occupancy · masked autoencoder · PyTorch*

2 · Perception & Trajectory Prediction

TopoDiffuser — Diffusion-Based Multimodal Trajectory Prediction [↗](#)

Reproduced the diffusion-based multimodal trajectory prediction pipeline with LiDAR-only inputs. Reached final displacement error under 1 m on the validation set, in line with the paper.

Stack: *Diffusion model · LiDAR · trajectory prediction · HD map topology · PyTorch*

ACTION100M — Hierarchical Video Action Annotation Pipeline [↗](#)

Built the Action100M annotation pipeline: temporal segmentation, hierarchical action labels (atomic to semantic), and export tooling for producing robot-learning pre-training data from raw video.

Stack: *Action100M · video annotation · hierarchical labelling · Python · data pipeline*

S2GO — Streaming Sparse Gaussian Occupancy Prediction [↗](#)

Reimplemented S2GO's camera-only semantic occupancy prediction on nuScenes to paper spec (sparse Gaussian queries, 6-layer transformer). Trained Stage-2 from scratch on a single GPU and benchmarked against the paper's no-pretraining ablation with a 16-class IoU harness.

Stack: *Occupancy prediction · 3D Gaussians · nuScenes · transformer · PyTorch*

CarPlanner — Auto-regressive Trajectory Planning via Large-Scale RL [↗](#)

Implemented the paper's three-stage planner in PyTorch: PointNet scene encoder, invariant-view transformer decoder, and 60-mode selector with rule-based trajectory selection.

Stack: *Trajectory planning · RL · transformer · autonomous driving · PyTorch*

Sensor2Sensor — Cross-Embodiment Sensor Synthesis (In Progress) [↗](#)

Implementing a minimal UNet-based cross-modality sensor-synthesis pipeline, adapted to the nuScenes 6-camera + LiDAR rig, to generate consistent data across differing sensor configurations.

Stack: *Sensor synthesis · UNet · nuScenes · LiDAR · PyTorch*

3 · Policy Learning & Vision-Language-Action Models

DoT — Decoder-only Transformer VLA Policy (Original) [↗](#)

Designed and trained a decoder-only transformer VLA that maps image observations and language instructions to 20-step action chunks. Achieves closed-loop task completion on the PushT benchmark.

Stack: *Decoder-only transformer · action chunking · VLA · PushT · PyTorch*

Pi0 — Flow Matching VLA: Deep Study & Extension [↗](#)

Studied Pi0 by reimplementing its core pieces — flow-matching action head, action chunking, and VLM-conditioned proprioception embeddings — and ran action-chunk-size ablations.

Stack: *Flow matching · action chunking · VLA · transformer · PyTorch*

Eureka — LLM Reward Design for Piano-Playing Dexterous Hand [↗](#)

Adapted NVIDIA's Eureka LLM reward-generation loop to a new task: a 16-DoF hand playing piano in MuJoCo. Wrote the environment, observations, and evaluation harness, and iterated GPT-4-generated rewards to produce sequential key presses.

Stack: *LLM reward design · GPT-4 · MuJoCo · dexterous hand · PPO · Eureka*

SPACeR — KL-Anchored PPO over a BC Policy (Waymo / GPUDrive) [↗](#)

Reproducing SPACeR's sim-agent training in GPUDrive on Waymo Open Motion: PPO with a per-agent KL anchor to a behaviour-cloned reference policy, rollout accumulation across 10K scenes, and the paper's training-dynamics evaluation.

Stack: PPO · KL regularisation · GPUDrive · Waymo Open Motion · PyTorch

Human Video → Dexterous Manipulation — 3DGS Pipeline (In Progress) [↗](#)

Building a pipeline that turns monocular human videos into robot manipulation data: HaMeR hand-pose extraction, object 6-DoF tracking, 3D Gaussian Splatting workspace reconstruction, and retargeting to a robot hand.

Stack: Hand pose estimation · 6-DoF tracking · 3D Gaussian Splatting · retargeting · dexterous manipulation

4 · Humanoid Control & Locomotion

Unitree G1 — RL Gait & Running (IsaacSim) [↗](#)

Wrote multi-term reward functions for G1 walking and running in Isaac Lab — foot-contact timing, torso stability, velocity tracking, energy. Extended the task to loco-manipulation: walking while an SO101 arm performs pick-and-place.

Stack: IsaacSim · Isaac Lab · PPO · reward shaping · Unitree G1 · SO101

Humanoid Digital Twin — 3D Gaussian Splatting World Model [↗](#)

Built a digital twin of the Unitree G1 in a photorealistic 3D Gaussian Splatting scene, with a world-model API that predicts future scene states from robot actions. Added cobblestone terrain to stress-test locomotion.

Stack: 3D Gaussian Splatting · World Model API · Python · Unitree G1

Motion Retargeting — MikuMiku Dance → Unitree Humanoid [↗](#)

Built a retargeting pipeline from MMD animation files to the Unitree URDF skeleton, solving joint remapping and bone-hierarchy mismatches — producing reference motions consumable by AMP-style imitation learning.

Stack: Motion retargeting · URDF · Python · MMD format · AMP

5 · LLM Optimisation

TurboQuant — KV Cache Compression (Original, Published to PyPI) [↗](#) [GitHub ↗](#)

Implemented and published TurboQuant, a near-optimal KV cache quantization library for HuggingFace transformers. Achieves ~8x memory reduction at 2 bits with inner-product distortion within ~2.7x of the Shannon theoretical limit.

Stack: KV cache quantization · Lloyd-Max · QJL · HuggingFace transformers · PyPI · Python

EXPERIENCE

Business Developer (Part-time) · Marino Corporation of India · Remote

Jan 2026 — Present

- ▶ Built and launched the company website (marinoindia.co.in) with end-to-end on-page SEO, driving organic search visibility and nationwide brand reach.
- ▶ Developed the digital product catalog, webshop, and registered vendor portal, streamlining B2B ordering and vendor onboarding.
- ▶ Grew inbound inquiry flow to ~\$40,000/month through digital marketing and lead-generation initiatives.

Mechanical Engineer · VSParticle · Delft, Netherlands

Aug 2024 — Apr 2025

- ▶ **Engineered** a multi-material nano-printer for novel material synthesis, co-leading hardware development from concept to functional prototype.
- ▶ **Rebuilt** gas recycling system using microcontrollers, sensors & valves, improving system reliability and reducing manual intervention.
- ▶ **Designed** custom Faraday enclosures and PCB layouts for electrocatalysis substrate holders, enabling new experiment configurations.

Research Assistant — HCI Lab · Northwestern University · Evanston, IL

Jul 2023 — Jan 2024

- ▶ **Prototyped** mechanical logic gates (AND, XOR, NOR) using nitinol shape-memory actuators, demonstrating fully mechanical computation without electronics.
- ▶ **Demonstrated** functional half/full adder circuits from 3D-printed parts and copper-plated contacts, validating a novel approach to physical computing.

Mechanical Engineer · Heliogen · Pasadena, CA

Aug 2021 — Apr 2023

- ▶ **Delivered** CAD/FEA (thermal & stress) design of a multi-ton 2 m solar particle receiver, driving design decisions through simulation-first methodology.
- ▶ **Analysed** 5M+ wind datapoints to model and visualise receiver particle loss, informing optimisation decisions for the heliostat field layout.
- ▶ **Secured** co-PI position on \$1.3M DOE project for high-temperature particle in-line mass flow sensor design.

Mechanical Design Engineer · EagleBurgmann Expansion Joints · Denmark

Oct 2014 – Aug 2019

- ▶ **Designed** large-scale expansion joints rated to 800°C & 200 bar per ASME/EN standards, delivering \$2M+ annual cost impact through design optimisation.

Powertrain Engineer (Formula SAE) · Ashwa Racing · Bangalore, India

Jan 2011 – Apr 2014

- ▶ **Designed** intake/exhaust system for a 600cc engine, achieving 40% gains in peak power and torque.